

Jeffrey Hu

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Education

University of Cambridge

PhD student

2025–present

Cambridge, UK

Supervisor: Professor Ayush Tewari.

Massachusetts Institute of Technology

Master of Science; Bachelor of Science

2014–2019

Cambridge, MA

Supervisor: Professor Antonio Torralba.

Work experience

Parallel Systems

Senior Software Engineer

Mar 2022–Apr 2024

Los Angeles, CA

- Led development of a real-time multitask computer vision network and data engine for an autonomous train perception system.
- Built a multitask network for 3D object detection, 3D track prediction, depth prediction, and bird's-eye-view segmentation.
 - Wrote a PyTorch library for training models with various backbones, task-specific heads, and loss functions.
 - Developed a representation for railroad track geometry by regressing 3D splines and classifying switch states.
 - Shipped multiple model iterations onto a Jetson using Rust, TensorRT, and DeepStream.
- Developed a 3D reconstruction and autolabeling pipeline for scalable ground-truth generation using Dagster and Kubernetes.
 - Used zero-shot, open-vocabulary foundation models for object detection, segmentation, tracking, and monocular depth estimation.
 - Aggregated predictions across time to reconstruct 3D scene representations as high-quality labels for training and evaluating the real-time network.
 - Built infrastructure to balance datasets, mine scenarios, and automatically generate test reports to support the regulatory safety case over time.
- Contributed to sensor calibration, onboard data collection, triggering, and upload infrastructure.

TuSimple

Research Engineer II

Jan 2020–Mar 2022

San Diego, CA

Amazon Prime Air

Software Engineer Intern

Jun 2016–Aug 2016

Seattle, WA

Research

Inria — GraphDeco

Research Engineer

Feb 2025–Sep 2025

Nice, France

- Worked with Professor George Drettakis on differentiable rendering with 3D Gaussian splats.
- Developed methods for intrinsic decomposition and editable physically-based reflections.

gsplat

Contributor

Apr 2024–Feb 2025

Contributed to an open-source Gaussian splatting library for 3D reconstruction and novel-view synthesis.

- Implemented 3DGS-MCMC for improved densification and bilateral grids for handling exposure changes.
- Combined Self-Organizing Gaussian Splats and spherical-harmonics k-means clustering for Gaussian splat compression.
- Added fisheye camera support, camera defocus blur modeling, and 2.5D Gaussian splatting for surface reconstruction.

Publications

Efficient Camera-Controlled Video Generation of Static Scenes via Sparse Diffusion and 3D Rendering

J. Chen, J. **Hu**, J. Lasenby, and A. Tewari

arXiv preprint, 2026.

Intrinsic Decomposition and Editing of 3D Gaussian Splats

A. Lanvin, J. **Hu**, S. Lucas, A. Bousseau, and G. Drettakis

I3D / Proceedings of the ACM on Computer Graphics and Interactive Techniques, 2026.

Adaptive Spatio-Temporal 3D Gaussian Splatting for Scenes with Oscillatory Motion

P. Tzathas, J. **Hu**, A. Meuleman, G. Cordonnier, and G. Drettakis

Eurographics / Computer Graphics Forum, 2026.

Editable Physically-based Reflections in Raytraced Gaussian Radiance Fields

Y. Poirier-Ginter, J. **Hu**, J.-F. Lalonde, and G. Drettakis

SIGGRAPH Asia, 2025.

gsplat: An Open-Source Library for Gaussian Splatting

V. Ye, R. Li, J. Kerr, M. Turkulainen, B. Yi, Z. Pan, O. Seiskari, J. Ye, J. **Hu**, M. Tancik, and A. Kanazawa

Journal of Machine Learning Research, 2025.

Miscellaneous

MIT Varsity squash — Captain

Programming: Python, PyTorch, C++, CUDA.

Interests: squash, cycling, skiing, hiking.